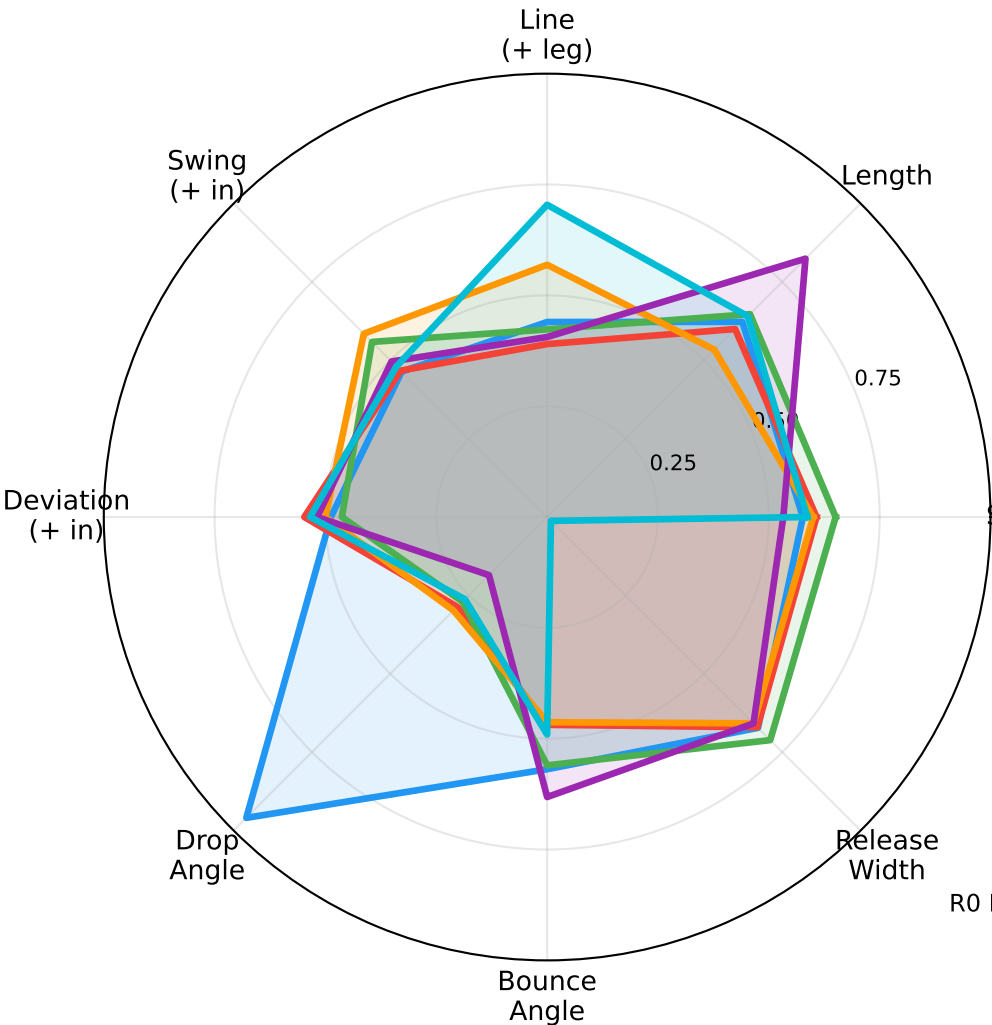


# MA Starc — delivery-type clusters

Cluster shapes (normalised)



Cluster centroids (within-bowler z-score)

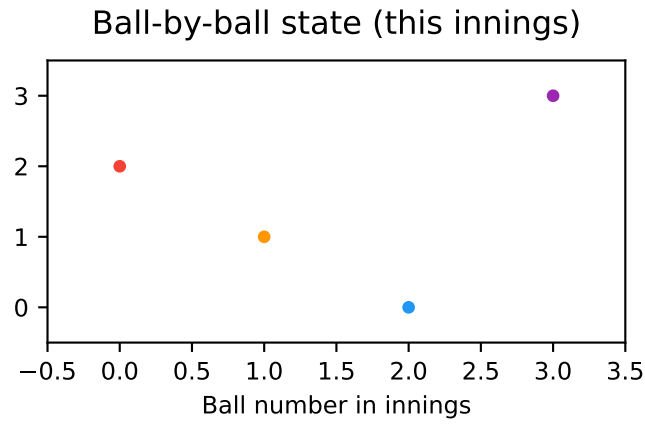
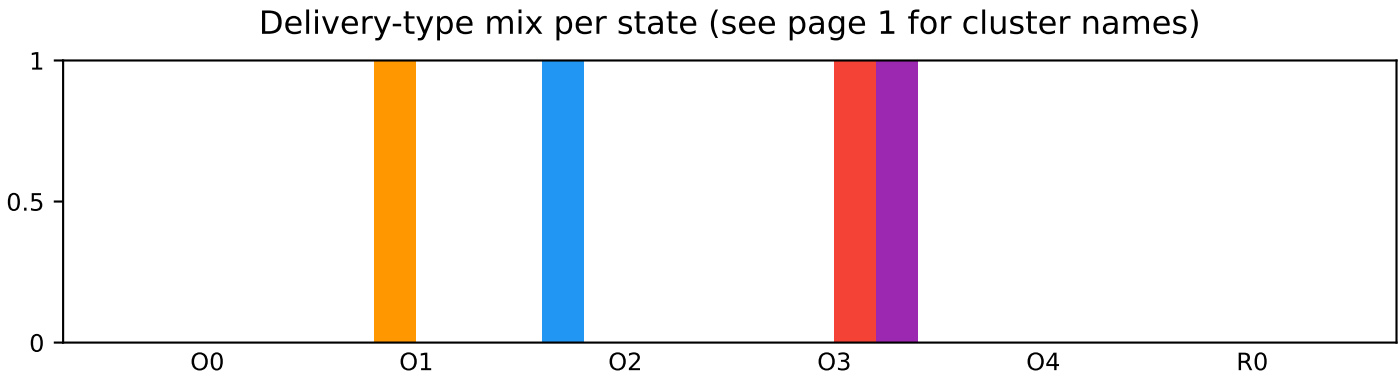
|                                             | Speed<br>(mph) | Length<br>(m) | Line<br>(bat adj) | Swing°<br>(bat adj) | Dev°<br>(bat adj) | Drop<br>Angle° | Bounce<br>Angle° | Release<br>Width |
|---------------------------------------------|----------------|---------------|-------------------|---------------------|-------------------|----------------|------------------|------------------|
| O0 Length, out-swinger<br>n=11, 9.1% wkt    | 87.9           | 6.3           | -0.3              | -0.2                | -0.2              | 11.2           | 8.6              | 0.4              |
| O1 Length, seam in<br>n=106, 4.7% wkt       | 88.5           | 5.9           | -0.4              | -0.2                | 0.3               | -11.1          | 7.2              | 0.4              |
| O2 Length, in-swinger<br>n=110, 2.7% wkt    | 89.4           | 6.8           | -0.3              | 0.8                 | -0.4              | -11.7          | 8.5              | 0.5              |
| O3 Full in-swinger<br>n=83, 1.2% wkt        | 88.4           | 4.8           | -0.0              | 1.1                 | -0.1              | -10.7          | 7.1              | 0.3              |
| O4 Bouncer<br>n=65, 4.6% wkt                | 87.0           | 9.8           | -0.3              | 0.1                 | 0.0               | -14.5          | 9.4              | 0.3              |
| R0 Round-the-wicket length<br>n=8, 0.0% wkt | 88.1           | 6.6           | 0.2               | -0.0                | 0.2               | -11.9          | 7.5              | -1.2             |

z-score

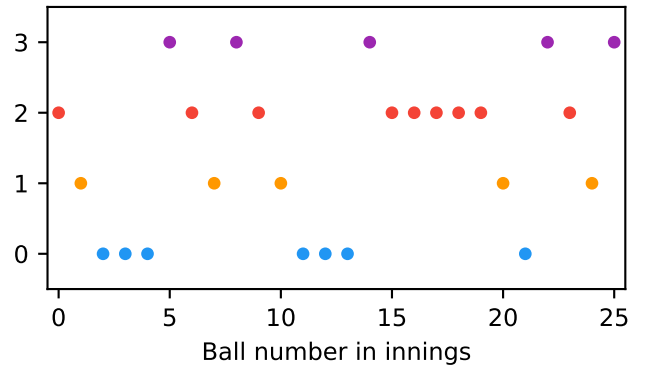
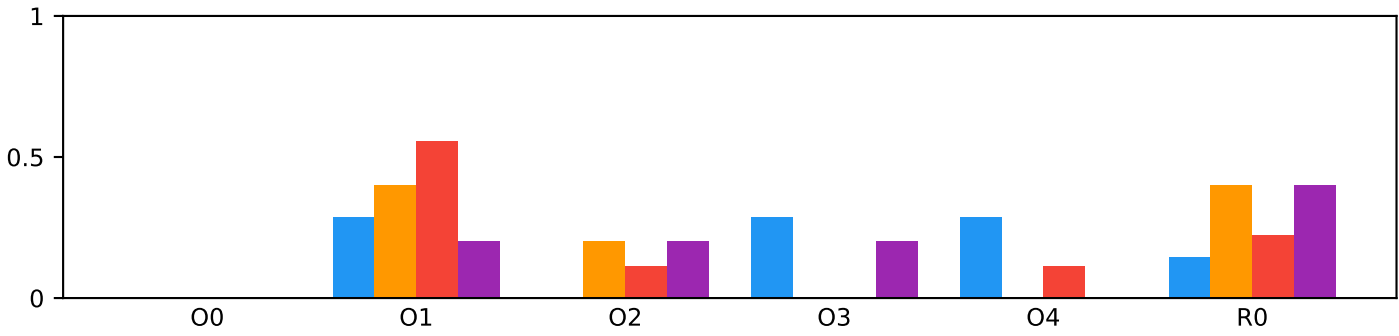
- O0 Length, out-swinger (n=11, 9.1% wkt)
- O1 Length, seam in (n=106, 4.7% wkt)
- O2 Length, in-swinger (n=110, 2.7% wkt)
- O3 Full in-swinger (n=83, 1.2% wkt)
- O4 Bouncer (n=65, 4.6% wkt)
- R0 Round-the-wicket length (n=8, 0.0% wkt)

★ = wicket

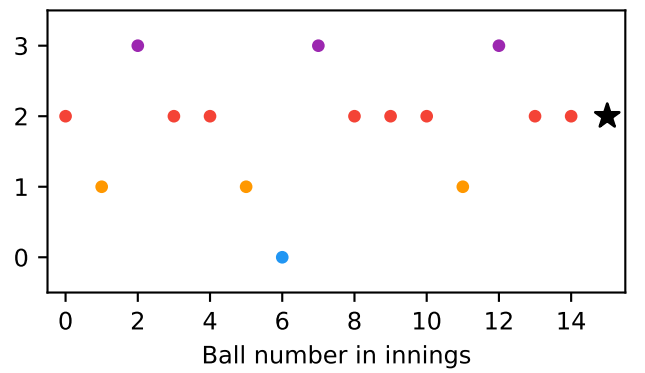
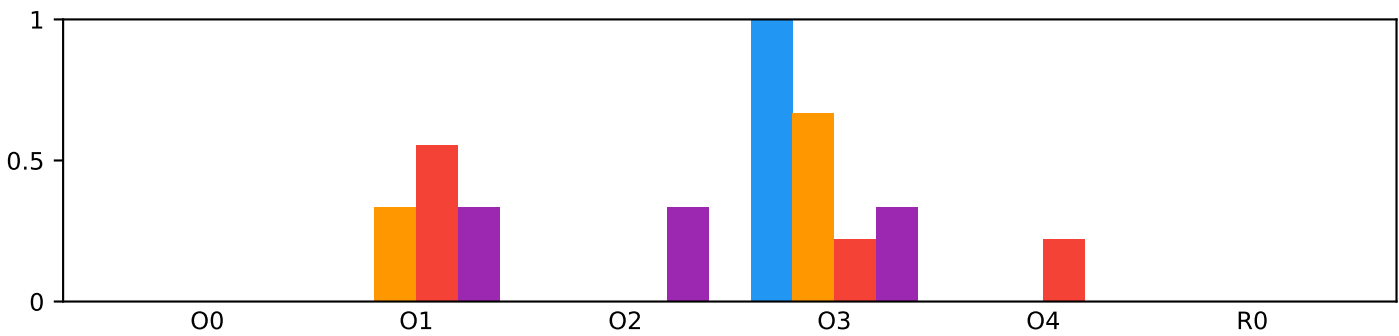
Innings 1  
2015-06-08  
Match ID 34022  
Innings ID 98265



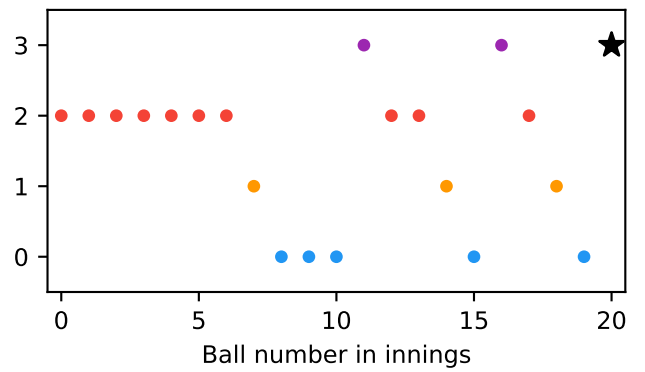
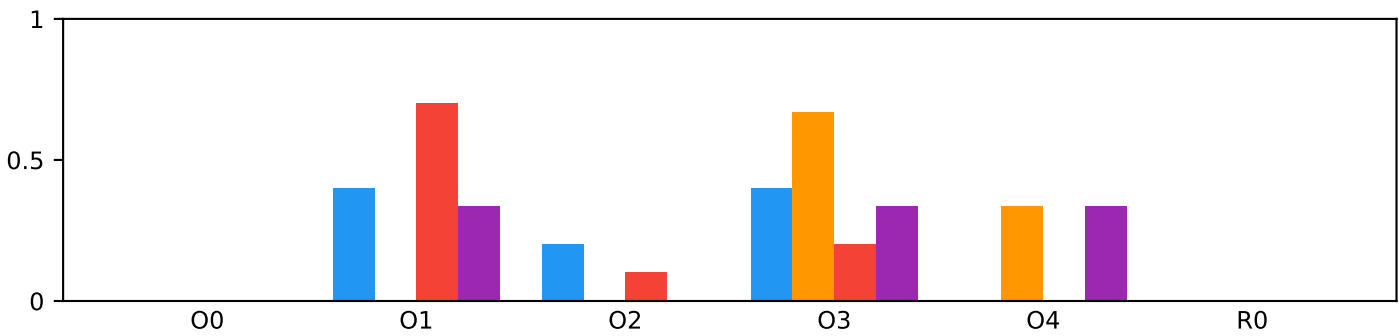
Innings 2  
2015-07-16  
Match ID 33897  
Innings ID 98254



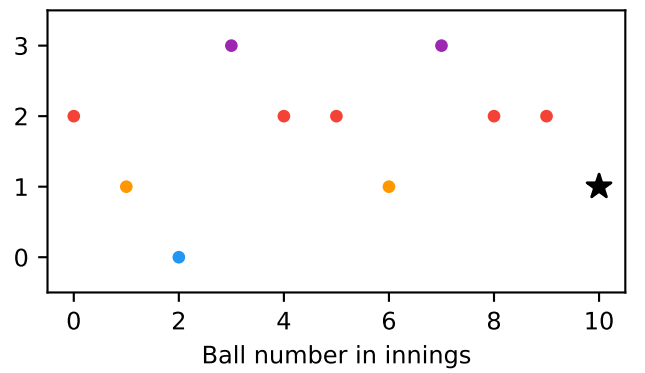
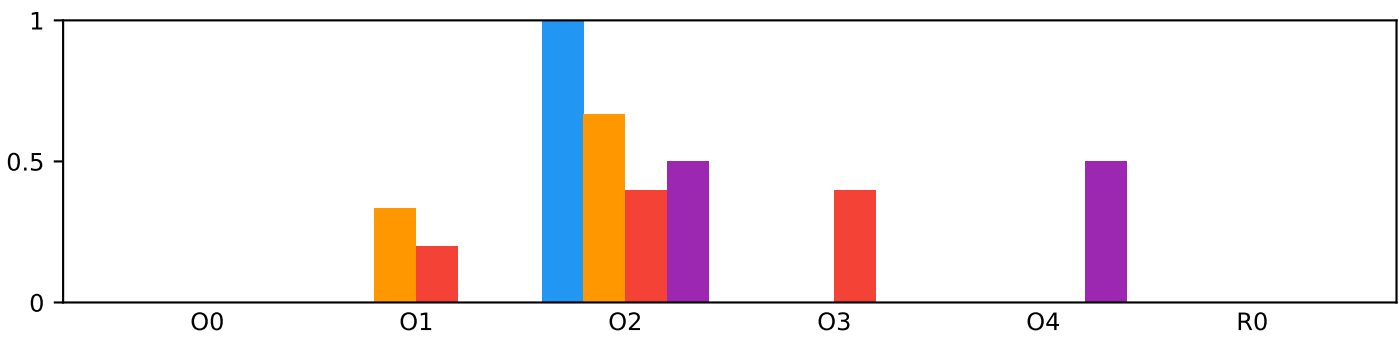
Innings 3  
2015-08-07  
Match ID 33817  
Innings ID 98249



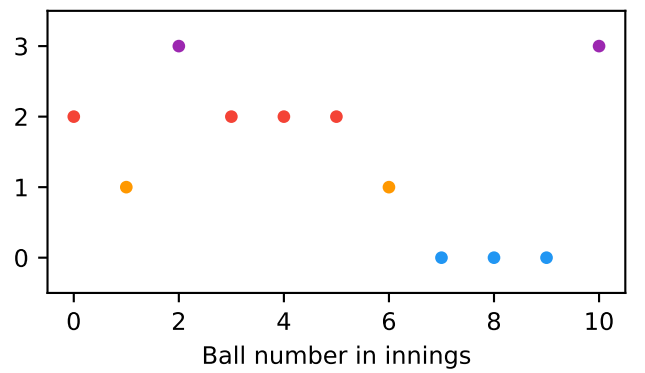
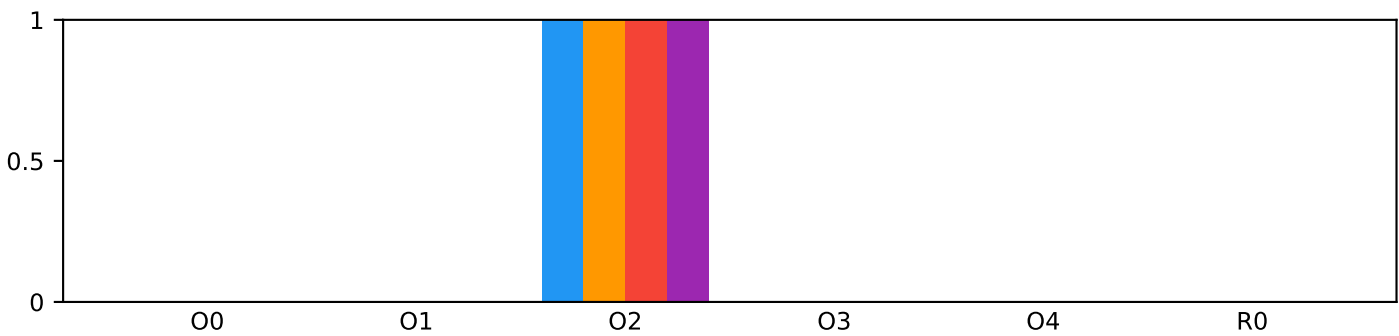
Innings 4  
2015-08-07  
Match ID 33817  
Innings ID 98251



Innings 5  
2019-04-09  
Match ID 44428  
Innings ID 137987



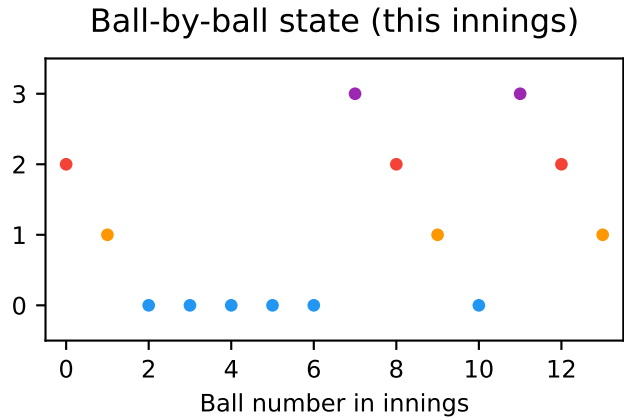
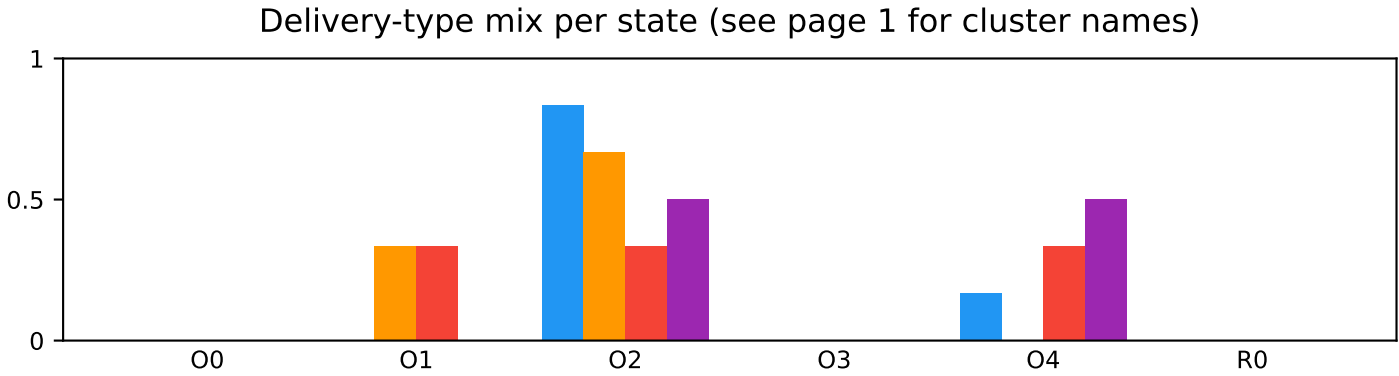
Innings 6  
2021-12-16  
Match ID 72171  
Innings ID 171961



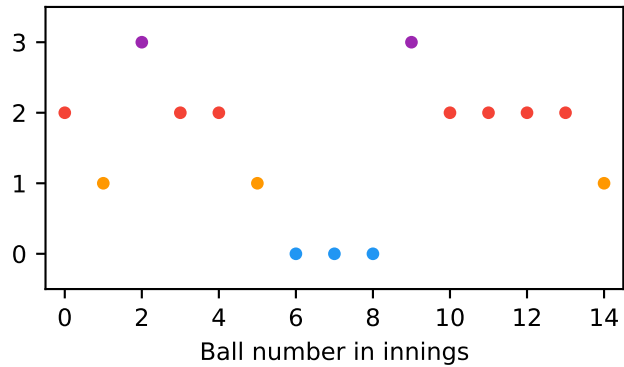
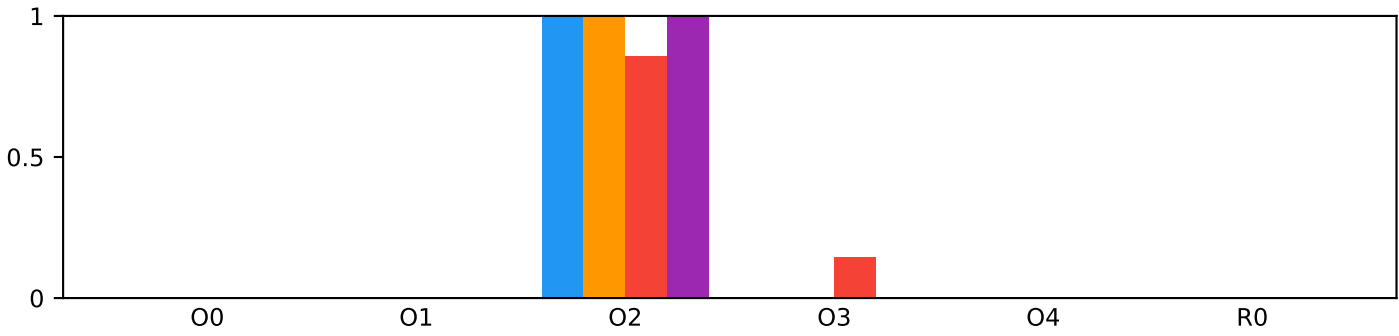
★ = wicket



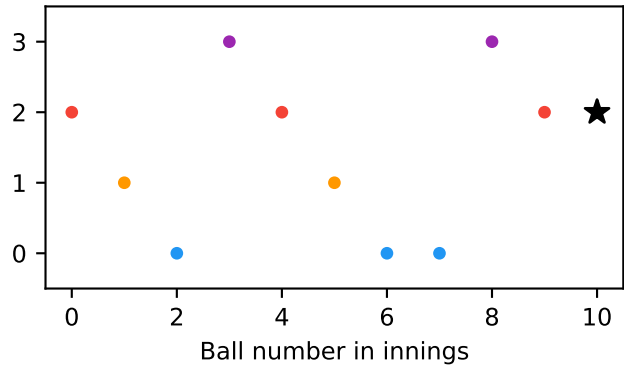
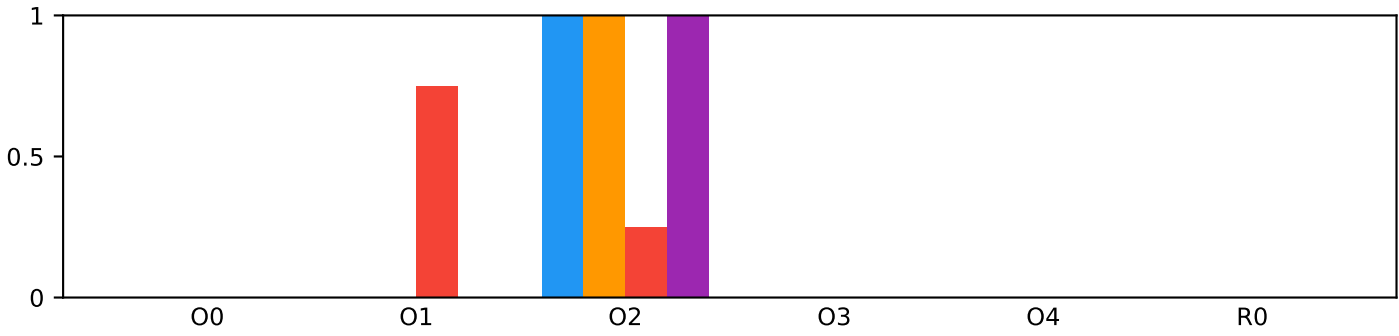
Innings 7  
2021-12-16  
Match ID 72171  
Innings ID 171967



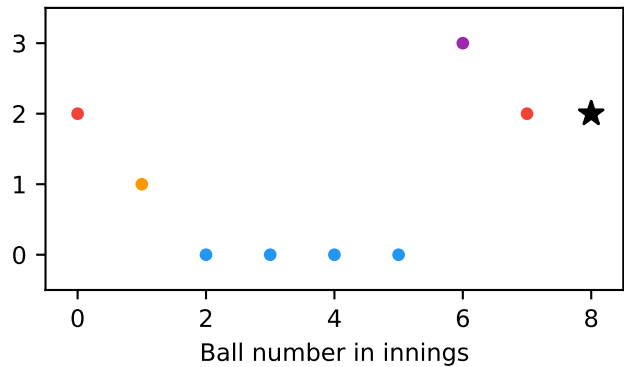
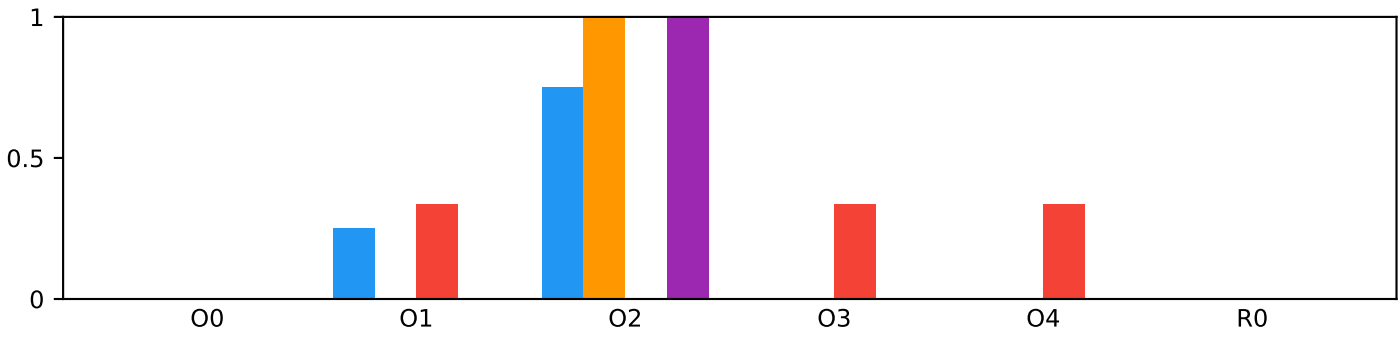
Innings 8  
2021-12-26  
Match ID 72172  
Innings ID 172187



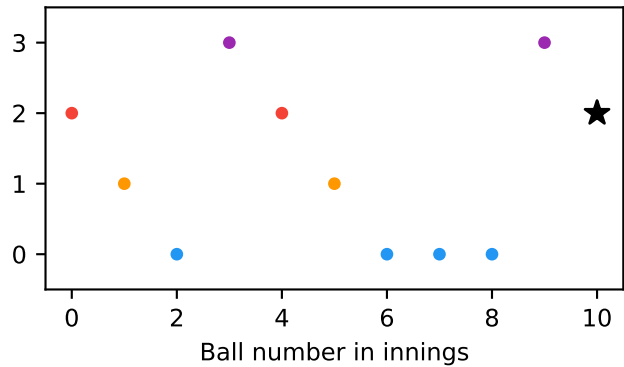
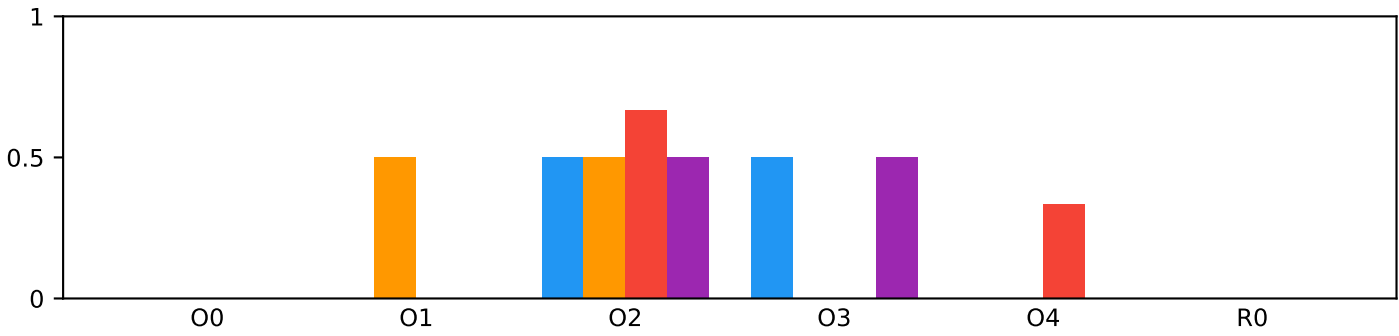
Innings 9  
2021-12-26  
Match ID 72172  
Innings ID 172202



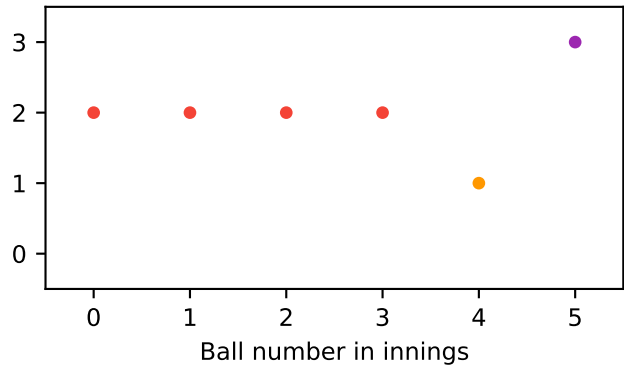
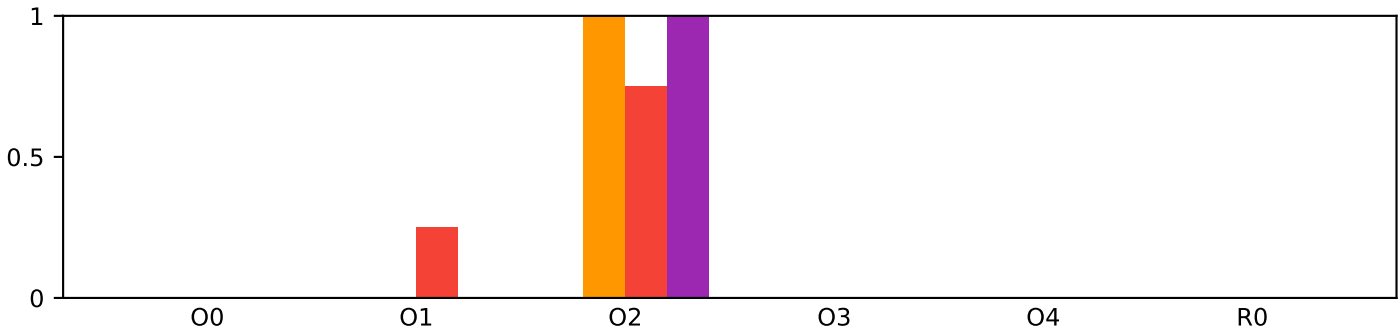
Innings 10  
2022-01-14  
Match ID 72174  
Innings ID 172495



Innings 11  
2022-01-14  
Match ID 72174  
Innings ID 172515



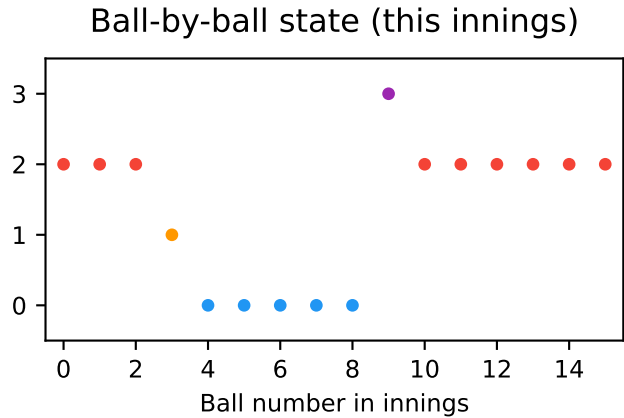
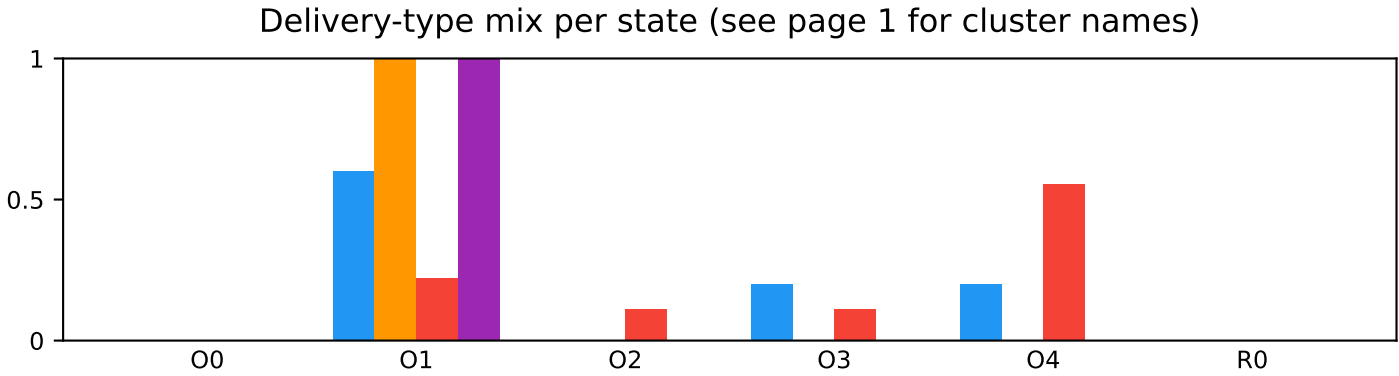
Innings 12  
2022-05-01  
Match ID 72173  
Innings ID 172281



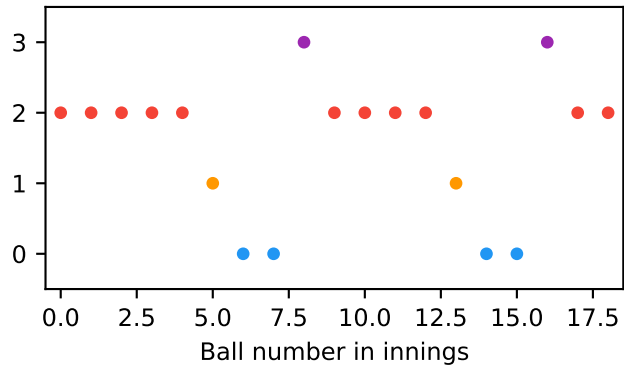
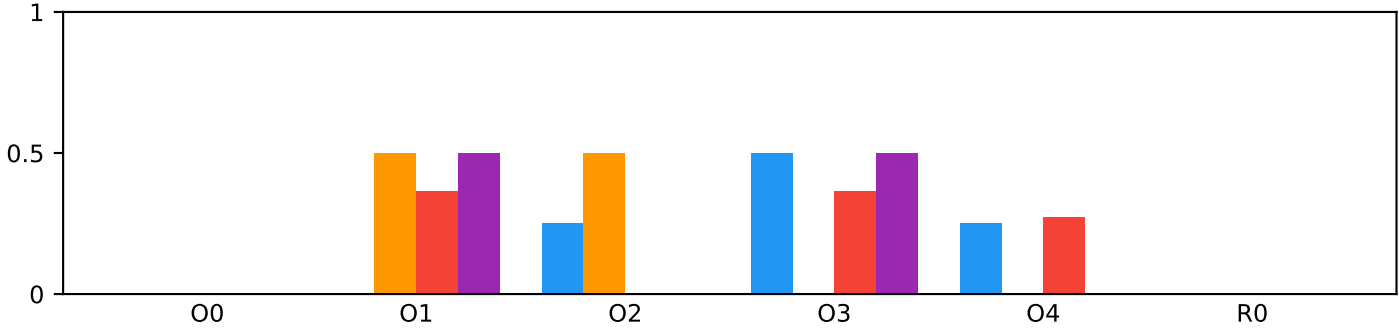
★ = wicket



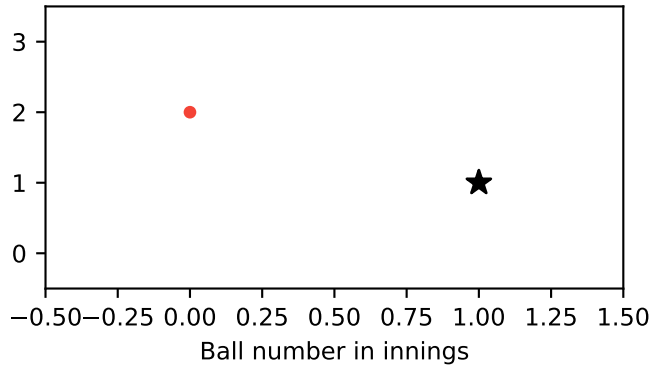
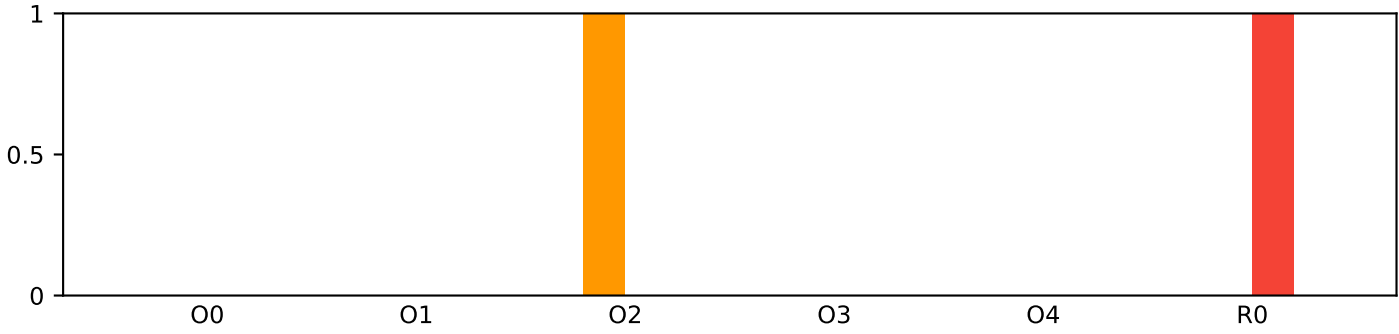
Innings 13  
2022-05-01  
Match ID 72173  
Innings ID 172414



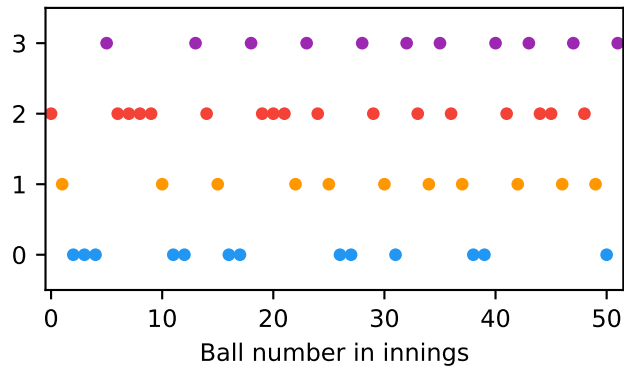
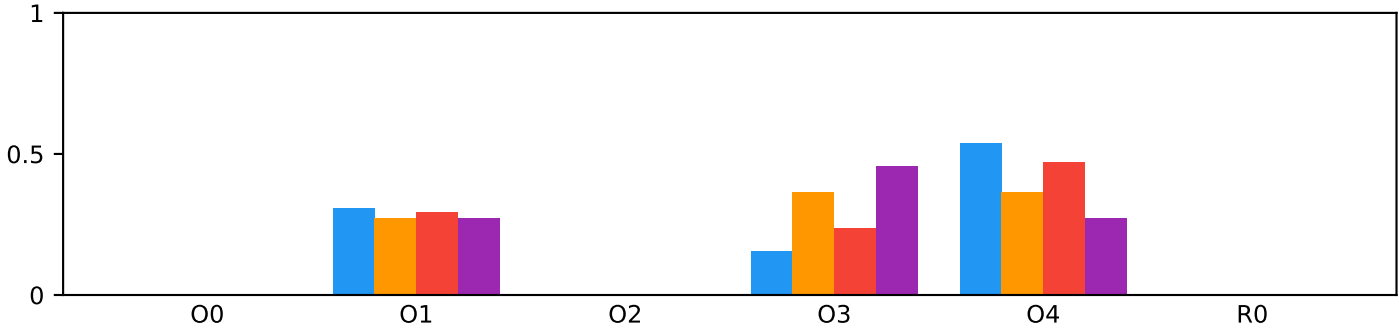
Innings 14  
2023-06-07  
Match ID 225571  
Innings ID 198391



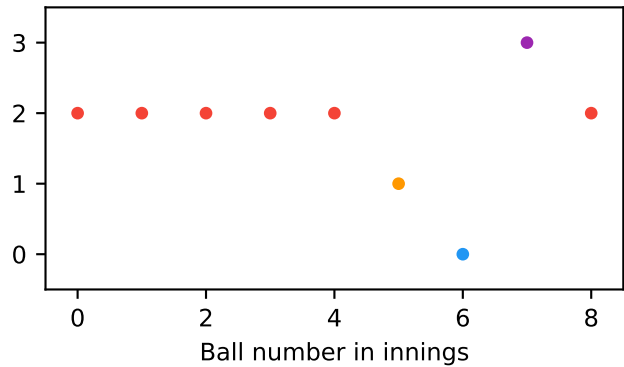
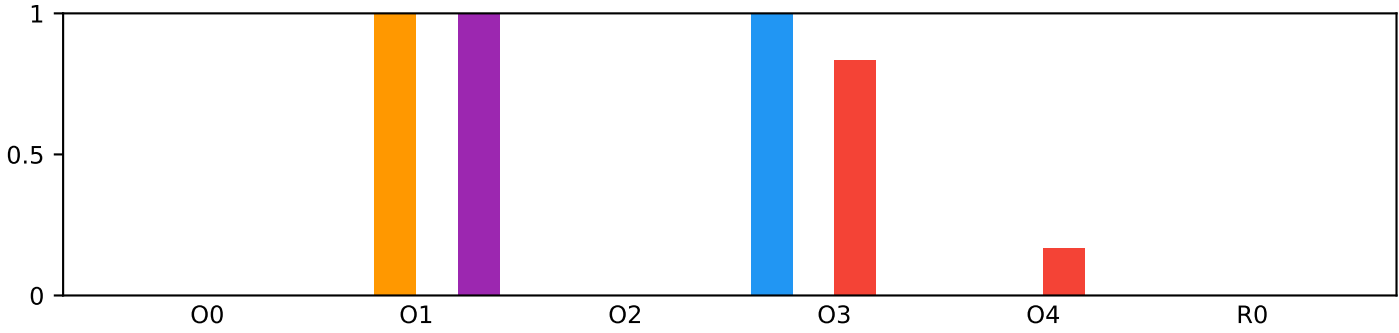
Innings 15  
2023-06-28  
Match ID 225570  
Innings ID 198060



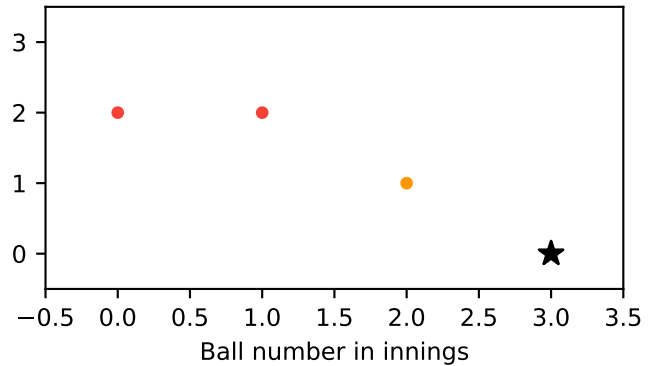
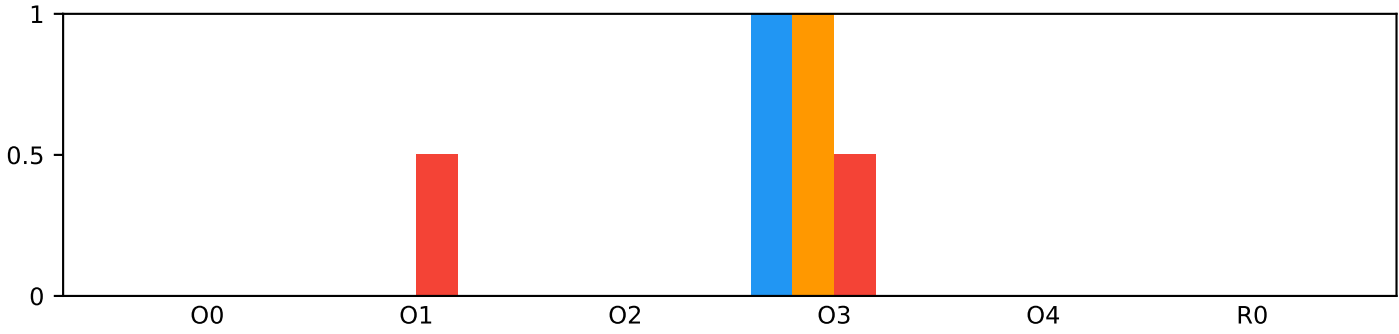
Innings 16  
2023-06-28  
Match ID 225570  
Innings ID 198114



Innings 17  
2023-07-19  
Match ID 225572  
Innings ID 199062

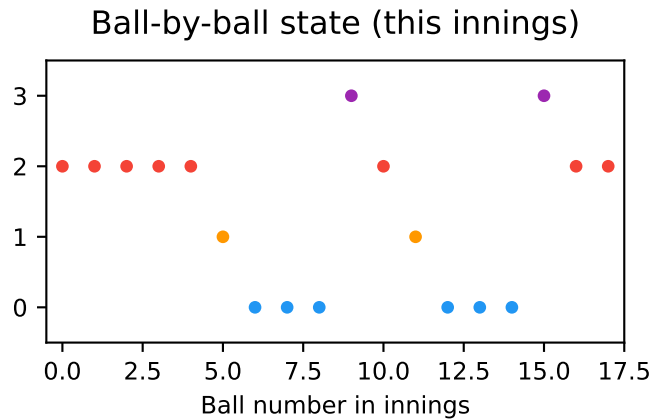
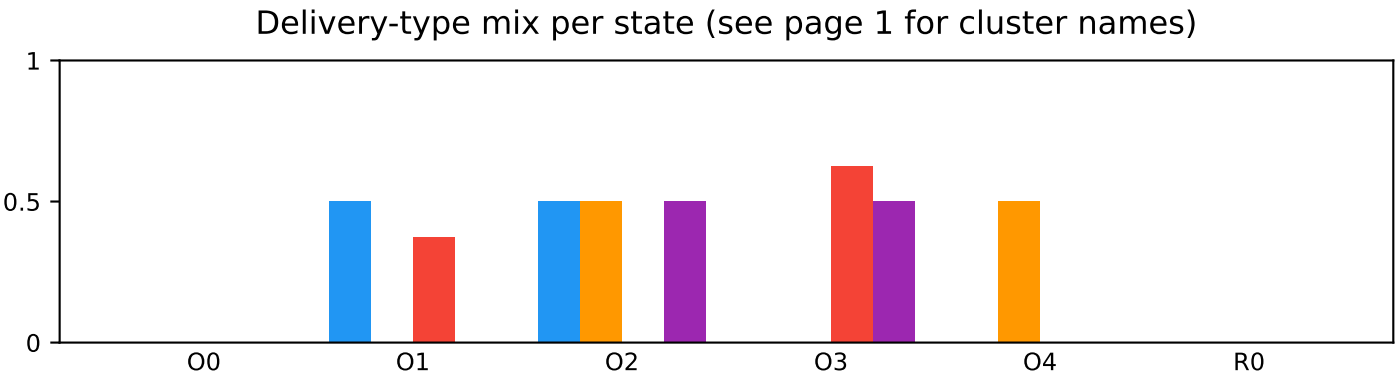


Innings 18  
2023-07-27  
Match ID 225573  
Innings ID 199426

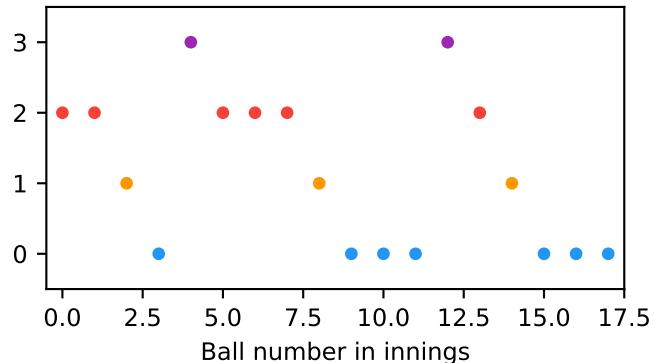
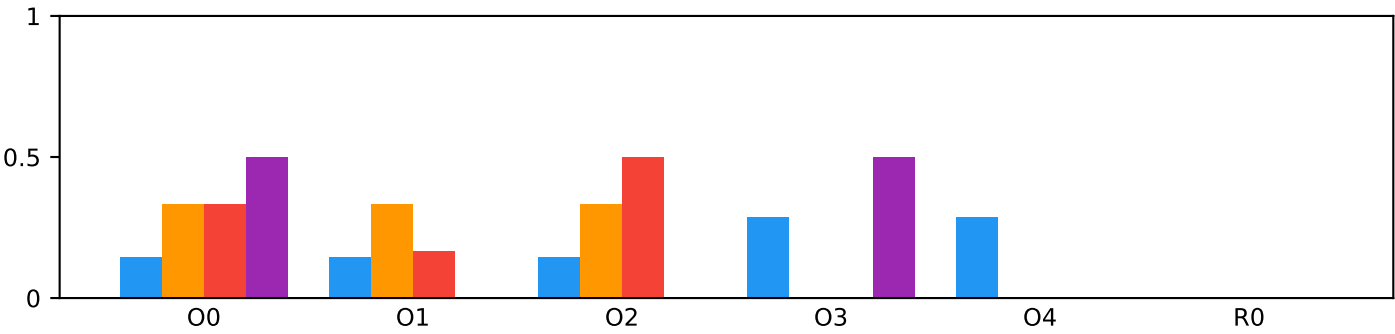


★ = wicket

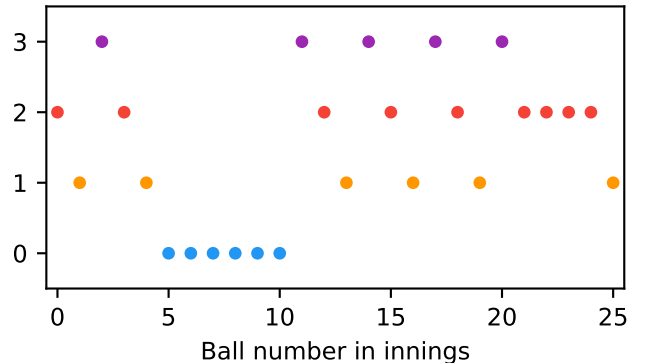
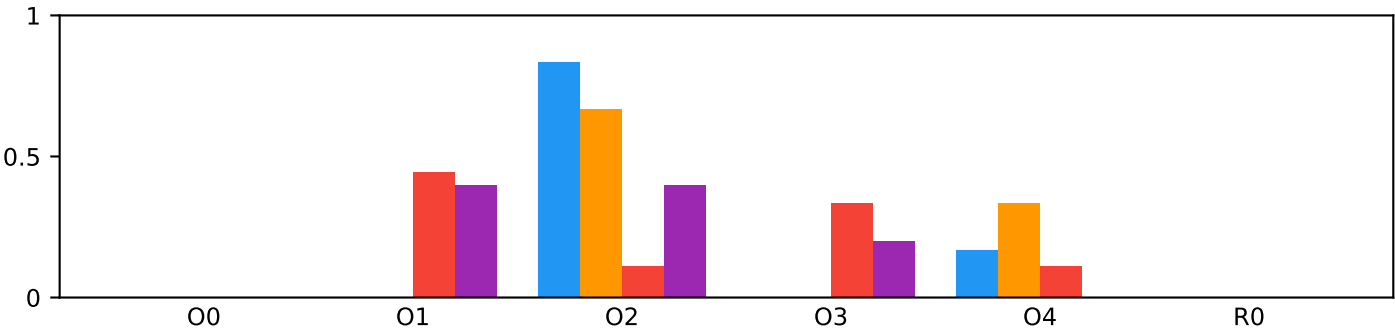
Innings 19  
2023-07-27  
Match ID 225573  
Innings ID 199520



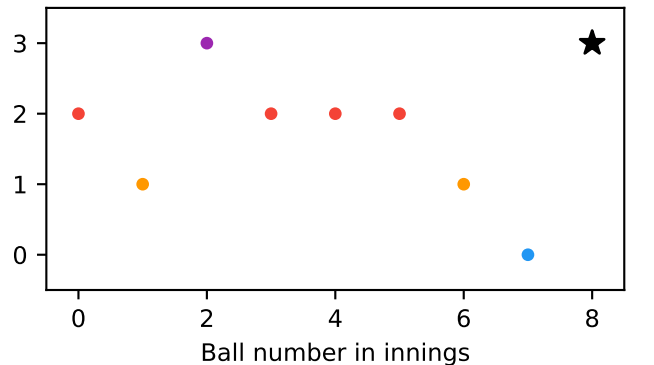
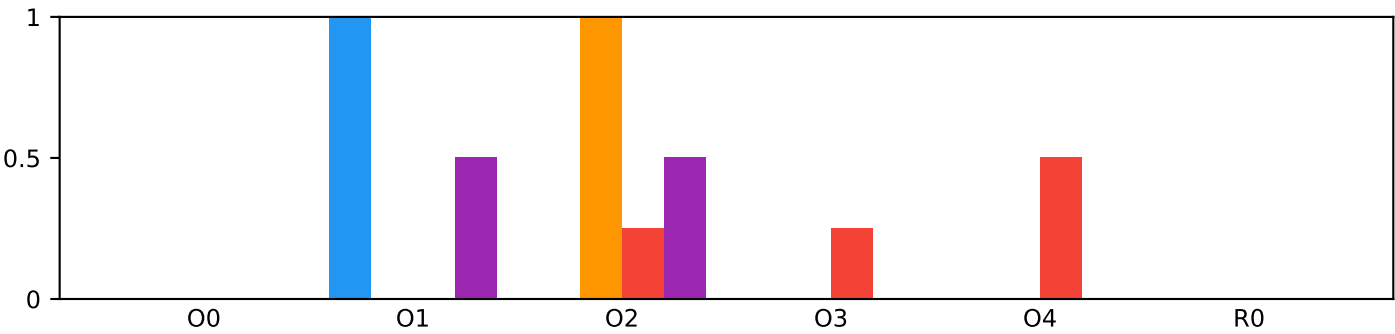
Innings 20  
2025-04-12  
Match ID 2571999  
Innings ID 254167



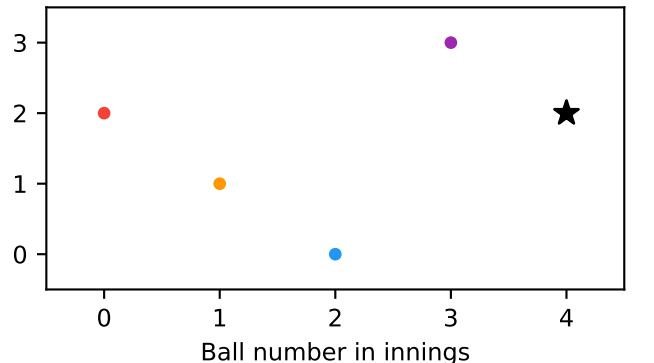
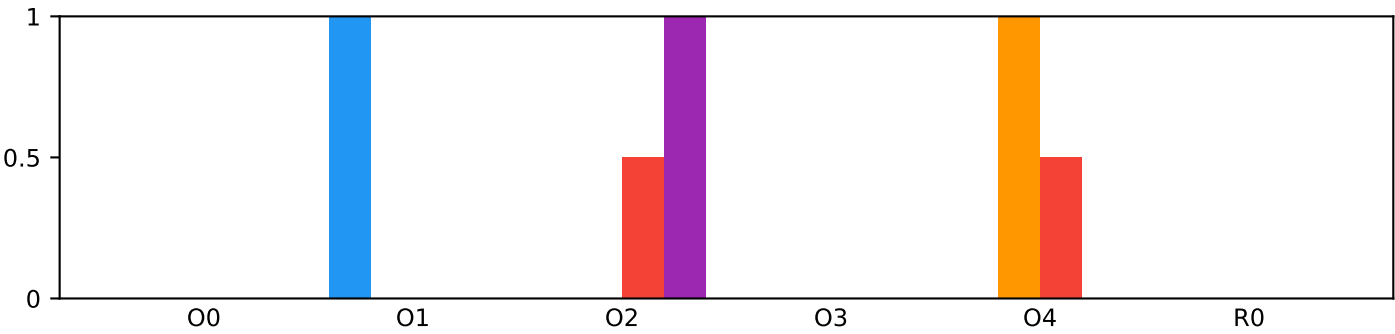
Innings 21  
2025-04-12  
Match ID 2571999  
Innings ID 254291



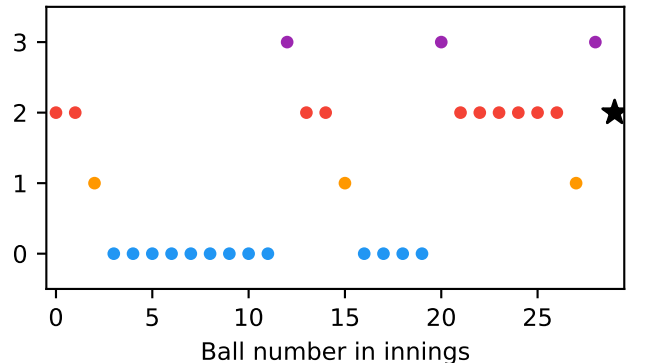
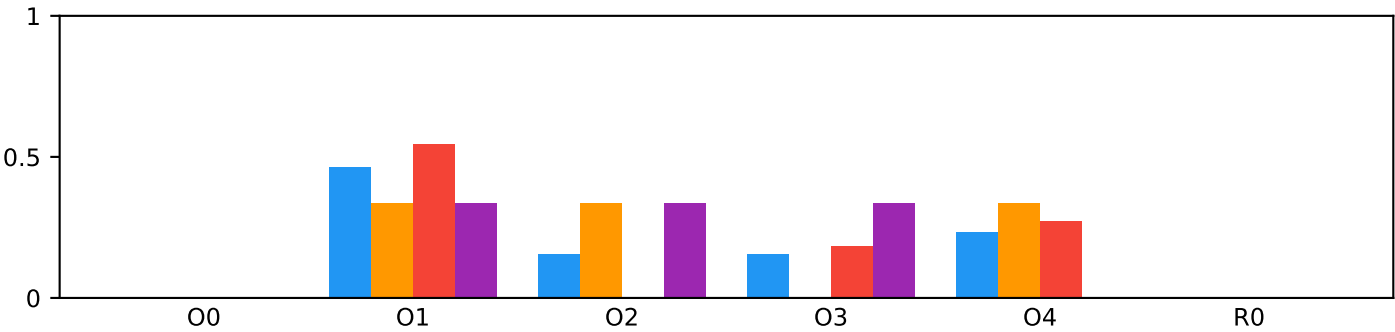
Innings 22  
2025-11-21  
Match ID 2571996  
Innings ID 253570



Innings 23  
2025-11-21  
Match ID 2571996  
Innings ID 253601



Innings 24  
2025-12-17  
Match ID 2571995  
Innings ID 254825

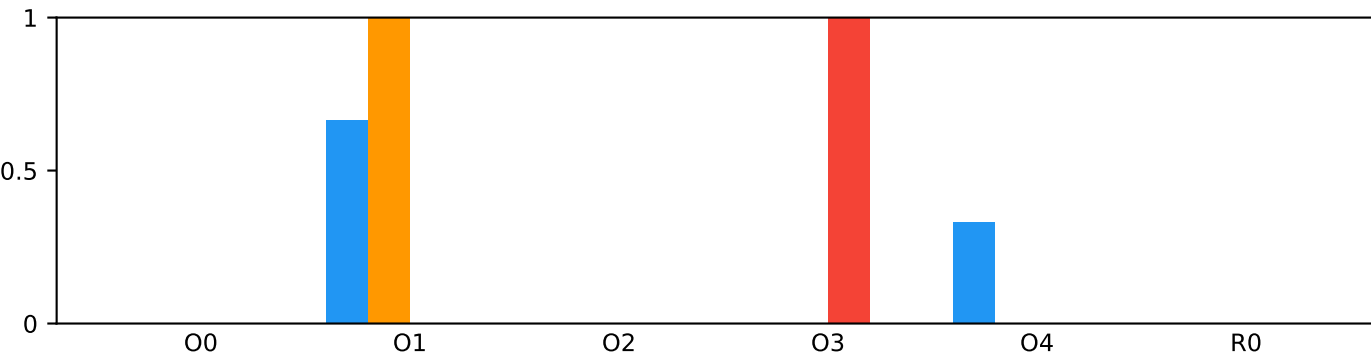


# MA Starc — per-innings strategy mix (page 5 of 5)

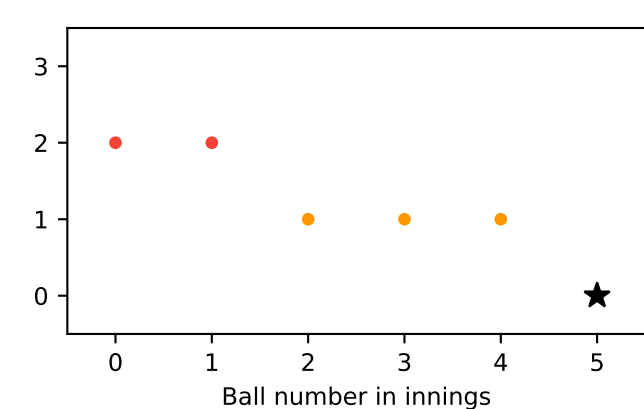
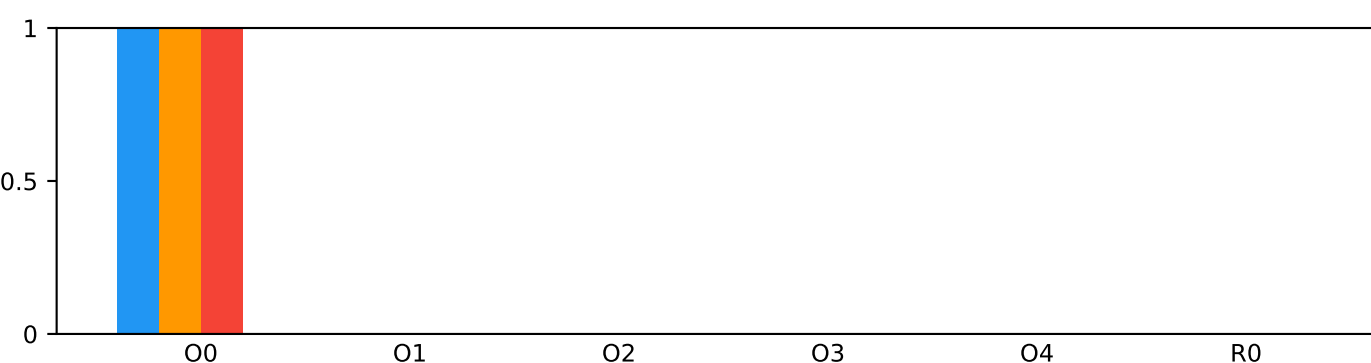
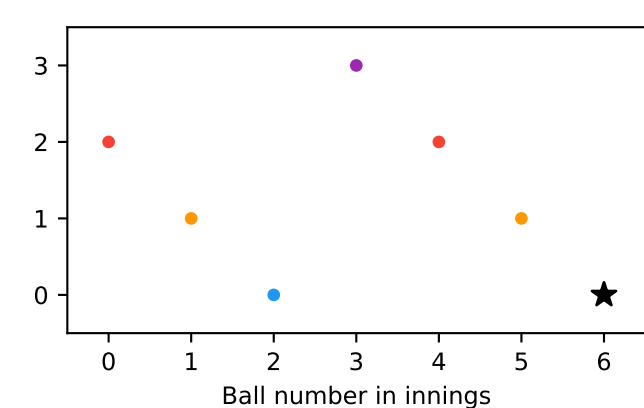
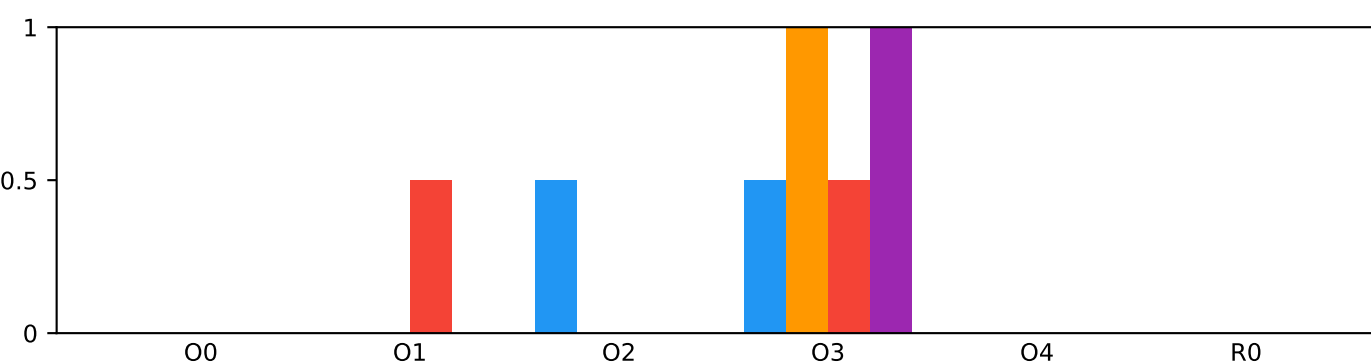
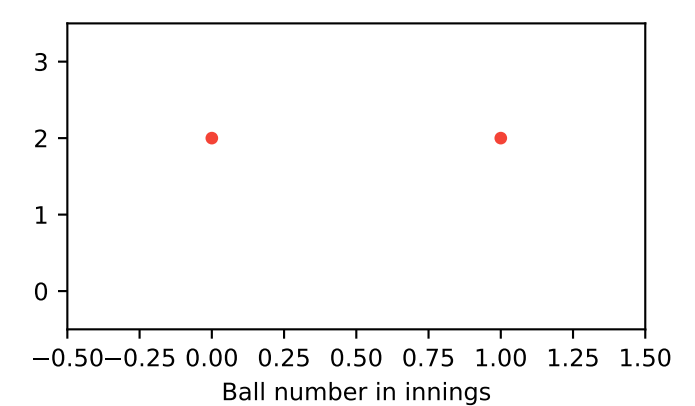
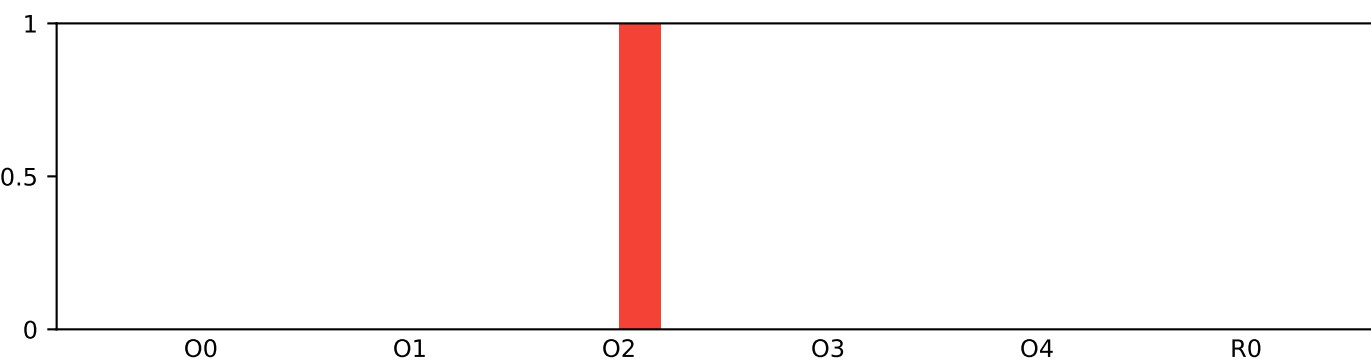
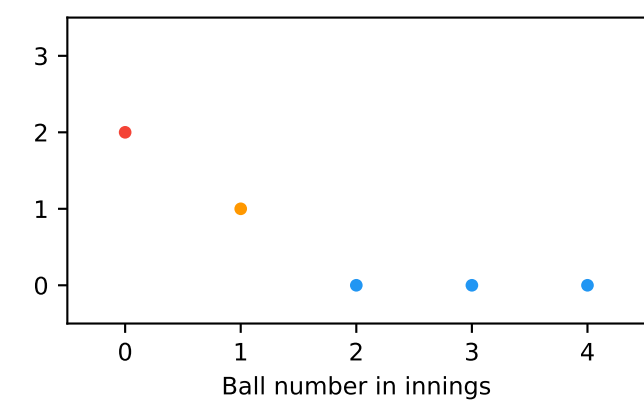
★ = wicket

State 0 State 1 State 2 State 3

Delivery-type mix per state (see page 1 for cluster names)



Ball-by-ball state (this innings)



Innings 25  
2025-12-17  
Match ID 2571995  
Innings ID 254890

Innings 26  
2025-12-26  
Match ID 2571997  
Innings ID 255074

Innings 27  
2025-12-26  
Match ID 2571997  
Innings ID 255101

Innings 28  
2026-04-01  
Match ID 2571998  
Innings ID 255414